Software Design Description 203

Sensors, PIR & Motor Subsystem

Overview

The sensor subsystem manages three PIR motion detectors, an AHT20 temperature/humidity sensor, a 28BYJ-48 stepper motor (via ULN2003 driver), and FPGA/camera enable pins. These are orchestrated by a single FreeRTOS task (`sensors_task`) that polls PIR inputs, reads environmental data, activates the camera and FPGA, and drives the stepper to aim at the triggered zone. Four source files implement the subsystem: `sensors_temp_humidity.c` (orchestrator), `aht20.c` (I2C sensor driver), `pir.c` (GPIO input), and `motor.c` (stepper sequences).

> **Hardware:** 3x PIR sensors | AHT20 on I2C0 (0x38) | 28BYJ-48 stepper + ULN2003 | FPGA/CAM enable GPIOs

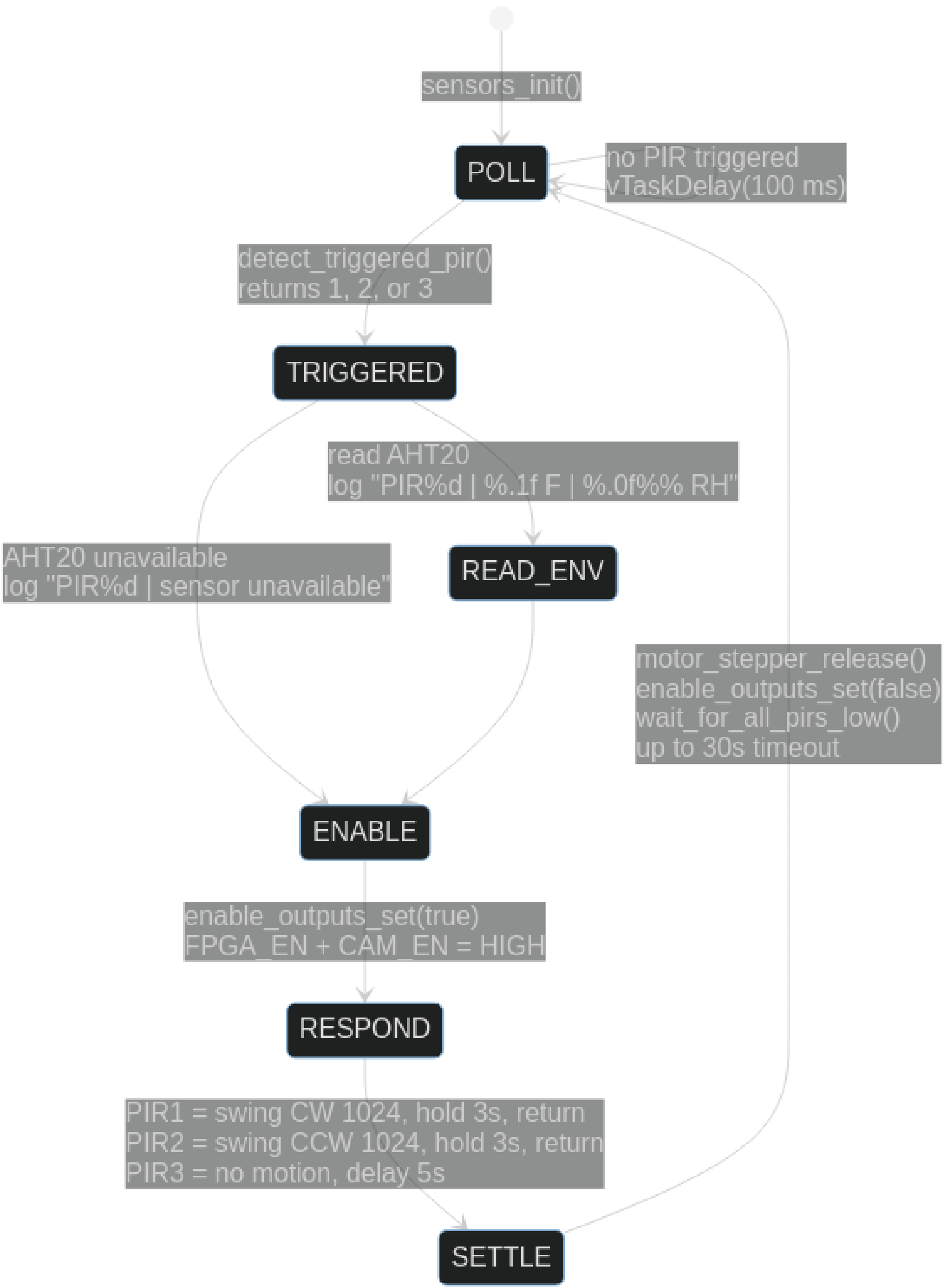
---

Table of Contents

- Sensor Task State Machine
- PIR Motion Detectors
- AHT20 Temperature/Humidity Sensor
- Stepper Motor
- FPGA & Camera Enable
- Pin Assignments
- Source Files
- References

Sensor Task State Machine

The `sensors_task` runs at low priority and implements a detect-respond-settle loop:


sensors_init()
POLL
no PIR triggered
vTaskDelay(100 ms)
detect_triggered_pir()
returns 1, 2, or 3
TRIGGERED
read AHT20
log "PIR%d | %.1f F | %.0f%% RH"
AHT20 unavailable
log "PIR%d | sensor unavailable"
READ_ENV
motor_stepper_release()
enable_outputs_set(false)
wait_for_all_pirs_low()
up to 30s timeout
ENABLE
enable_outputs_set(true)
FPGA_EN + CAM_EN = HIGH
RESPOND
PIR1 = swing CW 1024, hold 3s, return
PIR2 = swing CCW 1024, hold 3s, return
PIR3 = no motion, delay 5s
SETTLE

Response per PIR zone

PIR	GPIO	Direction	Motor Action	Total Active Time
PIR1	GPIO35	Left	CW 1024 steps, hold 3s, reverse 1024	~8s (ramp + hold + ramp)
PIR2	GPIO34	Right	CCW 1024 steps, hold 3s, reverse 1024	~8s
PIR3	GPIO4	Center	No motor, hold 5s	5s

1024 full steps on the 28BYJ-48 is approximately 90 degrees of rotation.

Settle phase

After responding, the task waits for all three PIR outputs to return LOW before re-entering the poll loop. This prevents retriggering on the same motion event. The wait has a 30-second timeout (`PIR_SETTLE_TIMEOUT_MS`) to avoid permanent stalls if a PIR stays asserted.

PIR Motion Detectors

pir.c

Minimal GPIO input driver. Configures three pins as inputs and provides `pir_read()`:

```
esp_err_t pir_init(int pir1, int pir2, int pir3);
bool      pir_read(pir_id_t pir);   // returns gpio_get_level()
```

The `sensors_temp_humidity.c` orchestrator does not use `pir.c` directly – it calls `gpio_get_level()` inline through `detect_triggered_pir()` for minimal overhead. `pir.c` exists as a standalone module for use by other subsystems.

GPIO configuration

PIR	GPIO	Input Mode	Pull-down
PIR1	GPIO35	Input only	Disabled (input-only pin, no pull available)
PIR2	GPIO34	Input only	Disabled (input-only pin, no pull available)
PIR3	GPIO4	Input/Output capable	Enabled

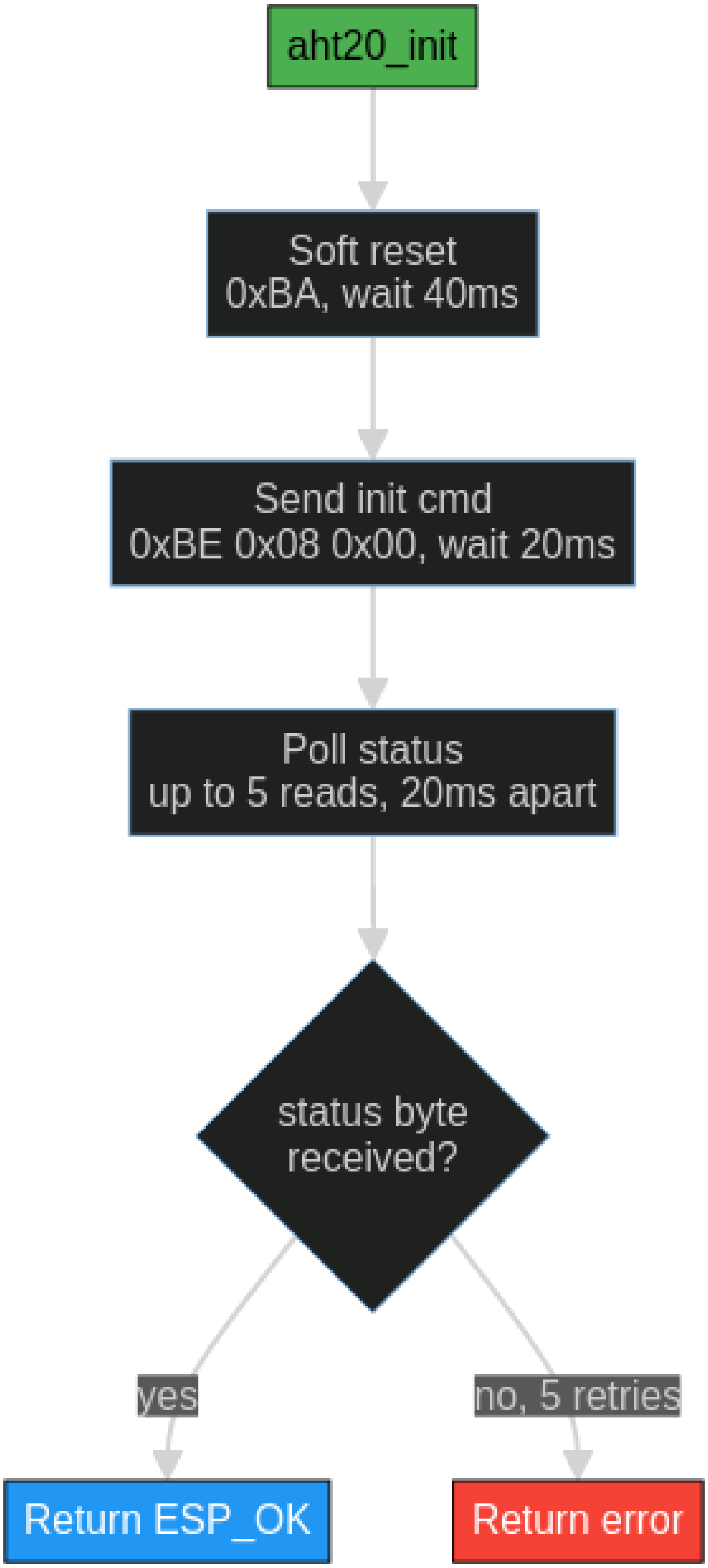
GPIO 34 and 35 are input-only pins on the ESP32 – they have no internal pull-up or pull-down capability. GPIO4 is a general-purpose pin with pull-down enabled.

AHT20 Temperature/Humidity Sensor

aht20.c

I2C driver for the AHT20 at address `0x38`. Uses `i2c_bus.c` for all bus access.

Init sequence



Measurement protocol

```
Write:  [0xAC, 0x33, 0x00]     Trigger measurement
Wait:   90 ms                   Conversion time
Read:   6 bytes                 [status, hum[19:12], hum[11:4], hum[3:0]|temp[19:16], temp[15:8], temp[7:0
```

Status byte bit 7 = busy. The driver polls up to 10 times (10 ms apart) waiting for busy to clear.

Raw-to-physical conversion

```
humidity (%) = hum_raw * 100.0 / 2^20
temperature (C) = temp_raw * 200.0 / 2^20 - 50.0
```

Where: - `hum_raw` = 20-bit value from bytes [1:3] upper nibble - `temp_raw` = 20-bit value from bytes [3] lower nibble + bytes [4:5]

The orchestrator converts Celsius to Fahrenheit for logging: `F = C * 9/5 + 32`.

Stepper Motor

motor.c

Driver for the 28BYJ-48 unipolar stepper motor through a ULN2003 Darlington array. Implements trapezoidal velocity ramping for smooth motion.

Step sequence

2-coil full-step pattern (highest torque):

Phase	IN1	IN2	IN3	IN4	Coils
0	1	1	0	0	AB
1	0	1	1	0	BC
2	0	0	1	1	CD
3	1	0	0	1	DA

Direction is determined by incrementing or decrementing the phase index (mod 4).

Wire map

The `wire_map[4]` field in `motor_stepper_t` remaps logical coil indices to physical GPIO outputs. The default map `{1, 0, 3, 2}` swaps coil pairs to fix a common wiring issue where the motor vibrates instead of rotating.

Trapezoidal ramp profile

`motor_stepper_move_ramped(m, steps, dir, start_ms, min_ms, ramp)`:

```
step delay (ms)
  ^
14|*
  | *
```

```
  |  *
  |   *
  |    *---------------------------*
2 |    |       cruise              |   *
  |    |                           |    *
  |    |                           |     *
  +----+---------------------------+------+----> step
  0   600                       steps-600  steps
```

- Ramp phase: delay linearly interpolated from `start_ms` (14) down to `min_ms` (2) over `ramp` steps (600)
- Cruise phase: constant `min_ms` delay
- Deceleration: mirror of acceleration ramp
- If total steps < 2*ramp, ramp is shortened to `steps/2`

High-level API

Function	Description
`motor_stepper_swing_cw(m, steps, hold_ms)`	CW ramp, hold, reverse ramp, release
`motor_stepper_swing_ccw(m, steps, hold_ms)`	CCW ramp, hold, reverse ramp, release
`motor_stepper_swing_and_release(m, steps, dir, hold_ms)`	Generic swing + return + release
`motor_stepper_release(m)`	De-energize all coils (prevents heating)

FPGA & Camera Enable

Two GPIO outputs gate power to the FPGA and camera subsystems:

Signal	GPIO	Purpose
FPGA_EN	GPIO2	Enable FPGA power/clock
CAM_EN	GPIO15	Enable camera module

Both are driven LOW on boot (`enable_outputs_init()`). When a PIR triggers, both are driven HIGH (`enable_outputs_set(true)`) for the duration of the motor response. After the motor sequence completes and coils are released, both are driven LOW again.

This allows the system to keep the FPGA and camera powered down during idle periods to conserve energy – relevant for battery/solar field deployment.

Pin Assignments

Signal	GPIO	Direction	Notes
PIR1	GPIO35	Input	Input-only pin, no pull
PIR2	GPIO34	Input	Input-only pin, no pull
PIR3	GPIO4	Input	Pull-down enabled
Stepper IN1	GPIO13	Output	ULN2003 input A
Stepper IN2	GPIO12	Output	ULN2003 input B
Stepper IN3	GPIO14	Output	ULN2003 input C
Stepper IN4	GPIO27	Output	ULN2003 input D
FPGA_EN	GPIO2	Output	FPGA power enable

Signal	GPIO	Direction	Notes
CAM_EN	GPIO15	Output	Camera power enable
I2C SDA	GPIO21	Open-drain	Shared with FPGA GPIO expander
I2C SCL	GPIO22	Open-drain	Shared with FPGA GPIO expander

Source Files

File	Lines	Purpose
`src/peripherals/sensors_temp_humidity.c`	225	Orchestrator: PIR poll, AHT20 read, motor + enable
`src/peripherals/aht20.c`	106	AHT20 I2C driver: init, measure, convert
`src/peripherals/pir.c`	33	PIR GPIO input: init, read
`src/peripherals/motor.c`	134	Stepper driver: step table, ramp, swing, release
`inc/peripherals/sensors_temp_humidity.h`	12	Public: sensors_init, sensors_task
`inc/peripherals/aht20.h`	22	Public: aht20_t struct, init, read
`inc/peripherals/pir.h`	16	Public: pir_id_t enum, init, read
`inc/peripherals/motor.h`	57	Public: motor_stepper_t, all motion functions

References

- SDD200 – DevKitV1 Overview
- SDD202 – I2C Bus & FPGA GPIO – I2C bus driver used by AHT20
- AHT20 Datasheet
- 28BYJ-48 Stepper Reference